

Advanced Diploma in Computer Applications

Detailed Chapter Notes & Worked Examples - Volume 2

This pack develops practical understanding through explanations, worked examples, lab tasks and review questions. Complete the matching Foundation Pack first.

Chapter	Topic
1	Office automation workflow
2	Relational data and reporting
3	Business accounts and GST
4	Integrated capstone

Chapter 1: Office automation workflow

Concept notes

A reliable office workflow moves from input to validation, processing, output and backup. Templates reduce repeated work; naming rules and version control prevent confusion.

Worked example

Create an admission register in Excel, generate a formatted confirmation letter in Word and prepare a five-slide monthly summary. Use StudentID as the common reference.

Lab assignment

Build the three files with 10 sample records and export final outputs to PDF.

Review questions

1. Why should a unique ID connect office records?
2. What must be checked before mail merge?
3. How does version control reduce mistakes?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 2: Relational data and reporting

Concept notes

A table stores one subject; each row is a record and each column is a field. A primary key identifies a record. Related tables avoid repeating the same details.

Worked example

Students(StudentID, Name, CourseCode) relates to Courses(CourseCode, Title, Fee). A query can combine both to show student, course and fee.

Lab assignment

Draw the two tables, enter 10 records and produce course-wise student totals.

Review questions

1. Define primary key.
2. Why avoid duplicate data?
3. What is the purpose of a query?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 3: Business accounts and GST

Concept notes

Computerized accounts follow the same debit-credit rules as manual accounting. Correct ledger groups and voucher types determine reliable reports. GST invoices need taxable value, rate and tax split.

Worked example

A local sale of Rs 10,000 at 18% GST produces CGST Rs 900 and SGST Rs 900; invoice total Rs 11,800.

Lab assignment

Create company, ledgers and five purchase/sales vouchers; verify P&L; and GST summary.

Review questions

1. When is IGST used?
2. Why is ledger grouping important?
3. What does trial balance test?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 4: Integrated capstone

Concept notes

A capstone converts a real requirement into data, process, output and controls. Start with scope; create a data dictionary; build; test normal and invalid cases; document backup and recovery.

Worked example

Institute fee system: student master, receipt entry, balance formula, due report, dashboard and printable receipt.

Lab assignment

Build a working prototype with 20 records and a one-page test report.

Review questions

1. What belongs in project scope?
2. Name two invalid test cases.
3. Why document recovery steps?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Final Applied Assignment

Create one original project that combines all four chapters of Advanced Diploma in Computer Applications. Use realistic but non-sensitive sample data. Submit planning notes, source files, final output, test evidence and a one-page reflection.

Assessment rubric

Area	Marks
Concept accuracy	25
Working practical output	30
Validation and testing	20
File organisation and presentation	15
Original explanation	10

Student declaration

I confirm that the submitted work is my own practice. I have not included passwords, OTPs, identity documents or confidential student records.

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